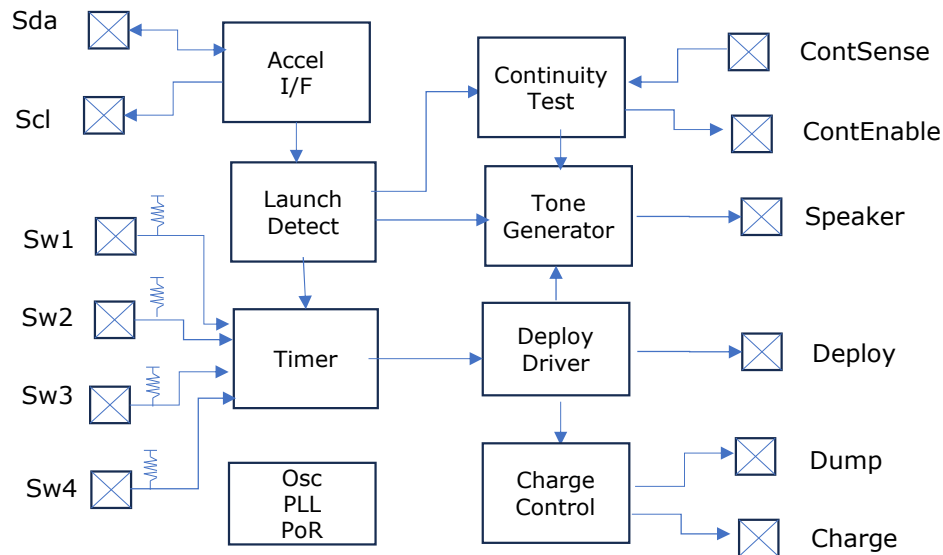


LDT Chip Block Diagram



Purpose

Launch-detect ASIC providing precise timed, safe and robust parachute deployment control.

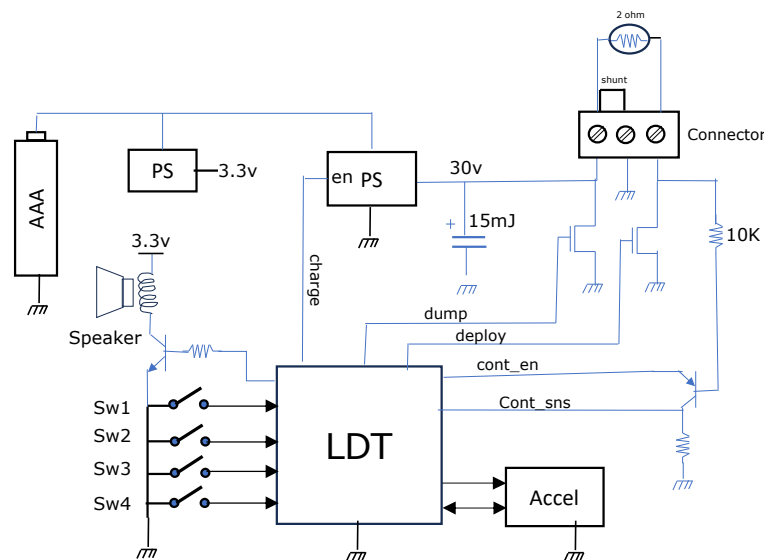
Features

- Precise launch-detect logic with programmable event timing
- Integrated continuity sensing for deployment circuit monitoring
- Automatic capacitor charge 0.5 seconds before deployment
- Direct dump and deployment Fet control outputs
- Safe-state defaults on reset or brownout events
- Built-in debounce filtering for launch-detect inputs
- Compact ASIC architecture minimizing external components
- Robust noise immunity for high-dynamic flight environments
- Configurable delay interval for multi-stage operations
- Low-power standby mode for extended pre-launch duration
- Fault-tolerant logic ensuring controlled event sequencing
- Clean separation of sensing, timing, and actuation domains
- Deterministic behavior independent of external processor timing

Typical Applications

- Deployment control with launch detect and precise event timing
- Staging event sequencing in compact avionics with integrated safety monitoring
- Airstart timing for secondary flight operations needing deterministic activation logic

System Diagram



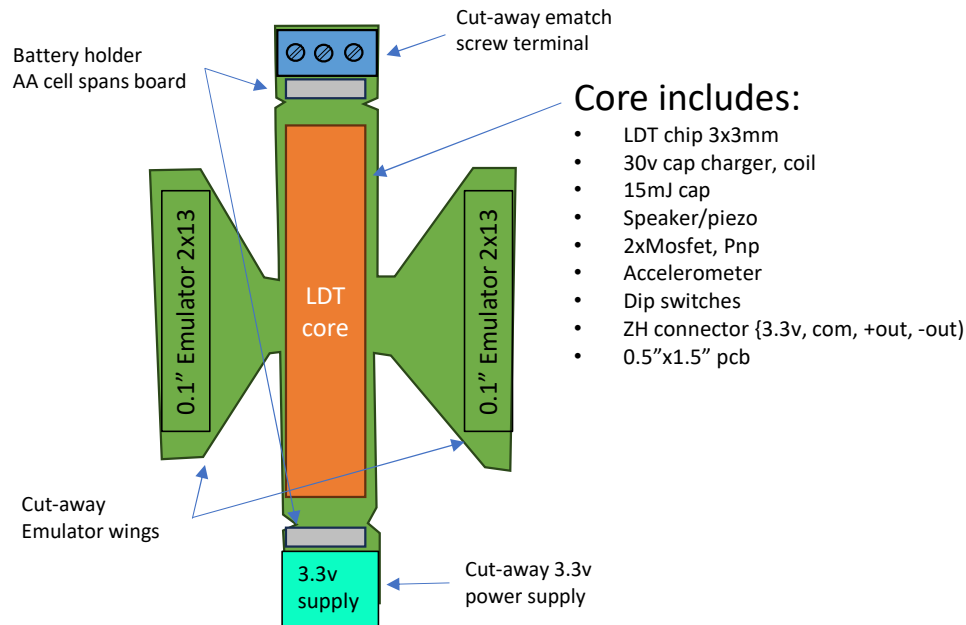
Description

Upon power-up, the device enters continuity mode, providing audible indicators of system health and deployment-circuit continuity. The user-selected delay interval is read from the configuration inputs. Launch is detected by sustained acceleration, either 1 G for 250 ms or 5 G for 180 ms, with the timer referenced to the start of acceleration. 500 msec before the scheduled deployment time, a 15 mJ capacitor is charged and then discharged precisely at the programmed delay, ensuring accurate and repeatable event timing. After deployment, the device enters post-event signaling mode and outputs a continuous warble tone aid for location.

System Behavior

- Continuity mode on power-up provides confirmation of system health and deployment-circuit integrity.
- Topology for external shunt to prevent accumulation of deployment energy, and while testing continuity continues independently.
- Launch detection uses sustained acceleration thresholds, starting the delay timer from the onset of measured motion.
- Event timing includes pre-deployment capacitor charge and precise discharge at the programmed delay, followed by continuous post-event signaling.

Development Board



Integration Value

- Provides complete standalone testing of all LDT system functions
- Enables rapid configuration and timing validation during development
- Supports flight-ready prototypes once emulation sections are removed

Purpose

The development board provides a self-contained platform for evaluating and integrating the LDT chip, combining power, sensing, configuration, and deployment interfaces in a compact form factor. A single AAA cell supplies the entire system, speaker, configuration switches, and screw-terminal deployment connections enabling complete functional testing without external hardware. The board supports full emulation of launch-detect, timing, capacitor-charge, and deployment behavior, and—with optional emulation sections removed—can serve as a flight-ready implementation for prototype or low-volume applications.

Features

- Single-cell AAA power architecture with onboard regulation
- Screw-terminal deployment interface for secure field connections
- Integrated speaker for continuity and status indication
- Onboard configuration switches and solder-ball options for delay setup
- Full launch-detect emulation using onboard accelerometer interface
- Supports capacitor discharge sequencing for deployment
- Breakaway emulation wings for optional flight-ready configuration
- Direct access to LDT chip I/O for development and debugging
- Compact, self-contained layout minimizing external hardware requirements
- Provides post-event warble tone for recovery and status identification
- Designed for bench evaluation, firmware integration, and prototype flight use